Personal Document & File Intelligence System

Novelty Verification: What's Actually New, Checked Against Existing Prior Art

Before finalizing novelty claims, targeted searches were run against each proposed contribution. The results required softening two of the three original claims. This document records exactly what was found and what remains defensible.

What Was Found (Existing Prior Art)

1. Document Availability Score

- A patent for **“automating formation and execution of a backup strategy”** computes a per-document **uniqueness score + importance score** to decide backup treatment — conceptually similar to combining signals into one document-level score.
- A separate patent computes **redundancy + significance** to set a document's “target availability level.”
- An **“Enterprise application high availability scoring”** patent combines backup/restore status, capacity, and documentation into a composite score.

Verdict: the idea of a composite, per-document score combining redundancy/backup/importance already exists in patented form. It is enterprise/backup-strategy oriented, not personal-user-facing, and none of it combines file-integrity into the same score — but the claim of “no precedent at all” could not be sustained.

2. Cross-Device Tamper Detection via Hash/Fingerprint Comparison

- A patent titled **“Efficient detection of ransomware attacks within a backup storage environment”** already samples bytes/fingerprints of backup files, compares them against a stored baseline, and issues an alert on mismatch — the same core mechanism proposed, applied to backup storage rather than peer personal devices.

Verdict: the detection mechanism (fingerprint baseline → compare → alert) is prior art. The distinguishing factor left is applying it peer-to-peer between personal devices with no server ever reading file content, rather than within centralized backup storage.

3. Risk-Prioritized Security Alerting

- Ransomware-response frameworks (CISA, Commvault, Acronis, Azure security guidance) all describe prioritizing response/recovery based on backup coverage and business impact — the general concept of “prioritize by what's actually at risk” is well established in enterprise incident response.

Verdict: the concept exists at the organizational/recovery-planning level. It has not been found applied to individual personal files for an individual consumer, with an automated per-document score driving real-time *alert urgency* rather than organizational recovery order.

Updated Novelty Table

Claim	Original Position	Position After Verification
Document Availability Score	No precedent found	Conceptually anticipated by backup-strategy scoring patents; not for personal, alert-facing use, and not combined with integrity in one score — narrower novelty, not zero
Cross-device hash-based tamper detection	No precedent found	Mechanism (fingerprint comparison → alert) exists in backup-storage patents; contribution is applying it peer-to-peer between personal devices with no server ever seeing content
Risk-prioritized alerting	No existing system ties alert severity to availability	General idea (prioritize by backup/impact) is standard in enterprise ransomware response; contribution is automating it per-document for an individual user via a computed score

What Is Actually Left as New — Precisely Stated

Not the mechanisms themselves (scoring, hash comparison, prioritization) — those all exist somewhere, mostly at enterprise/backup-infrastructure scale. What remains genuinely novel is narrower and more honest:

> ***"Repurposing existing enterprise/backup-strategy mechanisms — composite scoring, fingerprint-based tamper alerts, impact-based prioritization — into a single, unified, personal-user-facing system that runs entirely on-device with no centralized backup infrastructure, and that surfaces them to the user directly rather than automating backup-tier decisions behind the scenes."***

Implication for the Paper

- This is an **adaptation/integration novelty**, not an **invention novelty** — state this explicitly.
- Cite the identified patents and enterprise frameworks in the Related Work section as the closest prior art.
- Explain precisely how the proposed system differs: personal (not enterprise) scope, on-device execution, no centralized backup infrastructure, metadata-only sync, and user-facing alerting rather than backend automation.
- Do not claim these mechanisms as invented — a reviewer who finds this prior art independently (as this search just did) will treat an uncited overlap as a missed prior-art check, which damages credibility far more than citing it and explaining the distinction would.